

```
In [2]: import pandas as pd  
df=pd.read_csv(r'C:\Users\khara\Downloads\Fliipkart Sales Dataset.csv')  
df
```

Out[2]:

	OrderDate	OrderID	Delivery Date	CustomerID	Customer Age	Customer Gender	Location	Zo
0	31-05-2020	5136486	04-06-2020	230569424	57	M	Western	Zo
1	31-05-2020	5136457	12-06-2020	230568182	24	F	Western	Zo
2	31-05-2020	5136465	10-06-2020	230567584	31	M	Weija	Zo
3	31-05-2020	5136488	05-06-2020	230563788	62	F	Greater Accra	Zo
4	31-05-2020	5136449	10-06-2020	230563726	18	F	Weija	Zo
...	...	...	...	...	...	...	...	...
133498	04-04-2020	5115960	18-04-2020	230560355	24	F	Greater Accra	Zo
133499	04-04-2020	5115962	18-04-2020	230557477	27	F	Greater Accra	Zo
133500	04-04-2020	5115979	16-04-2020	230548894	37	M	Greater Accra	Zo

	OrderDate	OrderID	Delivery Date	CustomerID	Customer Age	Customer Gender	Location	Zo
133501	04-04-2020	5115981	22-04-2020	230547230	38	F	Ashanti	Zo

133502	04-04-2020	5115990	07-04-2020	230546677	45	M	Greater Accra	Zo
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133503 rows × 19 columns

```
In [8]: df['OrderDate'] = pd.to_datetime(df['OrderDate'])
df['Order Date_new'] = df['OrderDate'].dt.strftime('%d/%m/%Y')
df
```

Out[8]:

	OrderDate	OrderID	Delivery Date	CustomerID	Customer Age	Customer Gender	Location	Zo
0	2020-05-31	5136486	04-06-2020	230569424	57	M	Western	Zo
1	2020-05-31	5136457	12-06-2020	230568182	24	F	Western	Zo
2	2020-05-31	5136465	10-06-2020	230567584	31	M	Weija	Zo
3	2020-05-31	5136488	05-06-2020	230563788	62	F	Greater Accra	Zo
4	2020-05-31	5136449	10-06-2020	230563726	18	F	Weija	Zo
...	...	...	...	...	...	...	...	...
133498	2020-04-04	5115960	18-04-2020	230560355	24	F	Greater Accra	Zo
133499	2020-04-04	5115962	18-04-2020	230557477	27	F	Greater Accra	Zo
133500	2020-04-04	5115979	16-04-2020	230548894	37	M	Greater Accra	Zo

	OrderDate	OrderID	Delivery Date	CustomerID	Customer Age	Customer Gender	Location	Zo
133501	2020-04-04	5115981	22-04-2020	230547230	38	F	Ashanti	Zo
133502	2020-04-04	5115990	07-04-2020	230546677	45	M	Greater Accra	Zo

133503 rows × 20 columns

```
In [10]: df=df.drop(columns=['OrderDate'])
df
```

Out[10]:

	OrderID	Delivery Date	CustomerID	Customer Age	Customer Gender	Location	Zone	Delivery Type
0	5136486	04-06-2020	230569424	57	M	Western	Zone 3	Express
1	5136457	12-06-2020	230568182	24	F	Western	Zone 3	Standard Delivery
2	5136465	10-06-2020	230567584	31	M	Weija	Zone 2	Standard Delivery
3	5136488	05-06-2020	230563788	62	F	Greater Accra	Zone 1	Express
4	5136449	10-06-2020	230563726	18	F	Weija	Zone 2	Shipped from Abroad
...	...	...	...	...	...	...	...	...
133498	5115960	18-04-2020	230560355	24	F	Greater Accra	Zone 1	Standard Delivery
133499	5115962	18-04-2020	230557477	27	F	Greater Accra	Zone 1	Shipped from Abroad
133500	5115979	16-04-2020	230548894	37	M	Greater Accra	Zone 1	Standard Delivery

	OrderID	Delivery Date	CustomerID	Customer Age	Customer Gender	Location	Zone	Delivery Type
133501	5115981	22-04-2020	230547230	38	F	Ashanti	Zone 3	Shipped from Abroad
133502	5115990	07-04-2020	230546677	45	M	Greater Accra	Zone 1	Express

133503 rows × 19 columns

```
In [12]: print(df.duplicated())
```

```
0      False
1      False
2      False
3      False
4      False
...
133498   True
133499   True
133500   True
133501   True
133502   True
Length: 133503, dtype: bool
```

```
In [13]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 133503 entries, 0 to 133502  
Data columns (total 19 columns):  
#   Column                Non-Null Count  Dtype  
---  ---  
0   OrderID               133503 non-null int64  
1   Delivery Date         133503 non-null object  
2   CustomerID            133503 non-null int64  
3   Customer Age          133503 non-null int64  
4   Customer Gender       133503 non-null object  
5   Location              133503 non-null object  
6   Zone                  133503 non-null object  
7   Delivery Type         133503 non-null object  
8   Product Category     133503 non-null object  
9   SubCategory          133503 non-null object  
10  Product               133503 non-null object  
11  Unit Price            133503 non-null int64  
12  Shipping Fee          133503 non-null int64  
13  Order Quantity        133503 non-null int64  
14  Sale Price            133503 non-null int64  
15  Status                133503 non-null object  
16  Reason                34981 non-null  object  
17  Rating                133503 non-null int64  
18  Order Date_new        133503 non-null object  
dtypes: int64(8), object(11)  
memory usage: 19.4+ MB
```

```
In [14]: df.head()
```


Out[14]:

	OrderID	Delivery Date	CustomerID	Customer Age	Customer Gender	Location	Zone	Delivery Type	Pr Cat
0	5136486	04-06-2020	230569424	57	M	Western	Zone 3	Express	Fa
1	5136457	12-06-2020	230568182	24	F	Western	Zone 3	Standard Delivery	t
2	5136465	10-06-2020	230567584	31	M	Weija	Zone 2	Standard Delivery	t
3	5136488	05-06-2020	230563788	62	F	Greater Accra	Zone 1	Express	P
4	5136449	10-06-2020	230563726	18	F	Weija	Zone 2	Shipped from Abroad	Fa

In [15]: `df["Product Category"].describe()`

```
Out[15]: count          133503
unique           5
top      Health and beauty
freq           42447
Name: Product Category, dtype: object
```

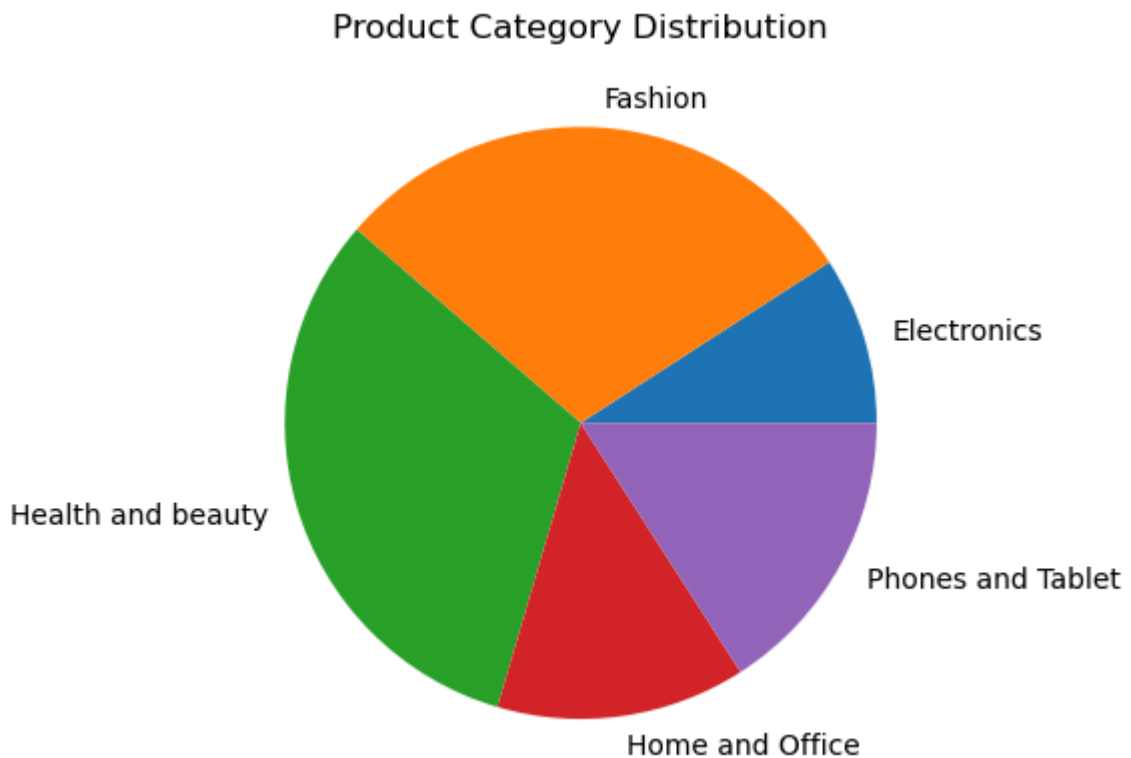
In [17]: `df["SubCategory"].describe()`

```
Out[17]: count          133503
unique          16
top      Vitamins & Dietary Supplements
freq           15089
Name: SubCategory, dtype: object
```

In [18]: `df["Product"].describe()`

```
Out[18]: count      133503
         unique         44
         top      6030 3.1 Bluetooth Home Theatre With Remote Co...
         freq      3069
         Name: Product, dtype: object
```

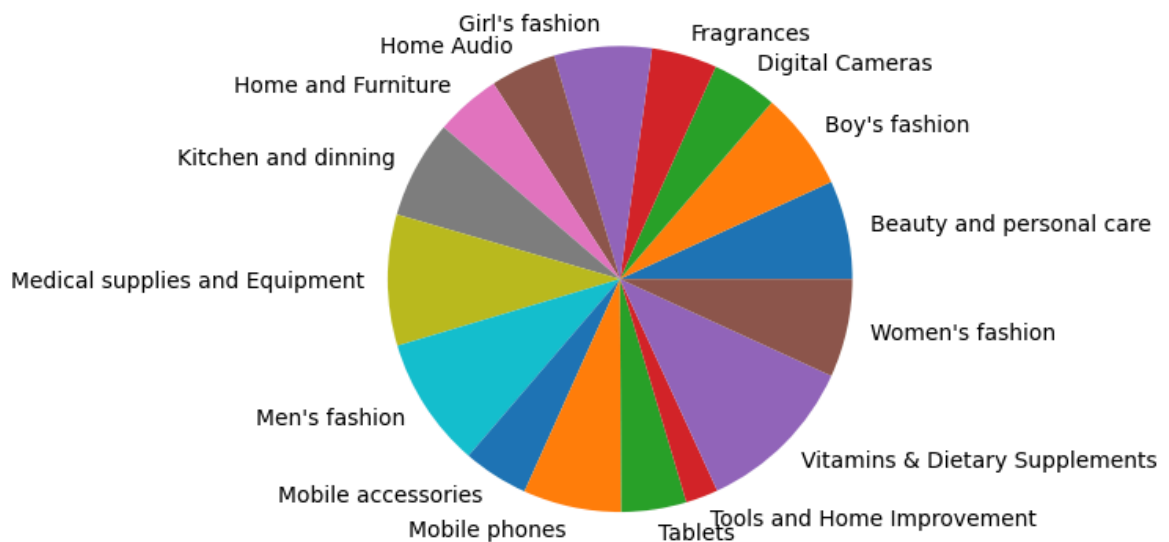
```
In [4]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
df = pd.read_csv(r'C:\Users\khara\Downloads\Fliptkart Sales Dataset.csv')
categories, counts = np.unique(df['Product Category'], return_counts=True)
plt.pie(counts, labels=categories)
plt.title('Product Category Distribution')
plt.show()
```



```
In [11]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
df = pd.read_csv(r'C:\Users\khara\Downloads\Fliptkart Sales Dataset.csv')
ratings, counts = np.unique(df['Rating'], return_counts=True)
plt.bar(ratings, counts, color='skyblue')
plt.xlabel('Customer Rating')
plt.ylabel('Number of Orders')
plt.title('Customer Rating Distribution')
plt.show()
```

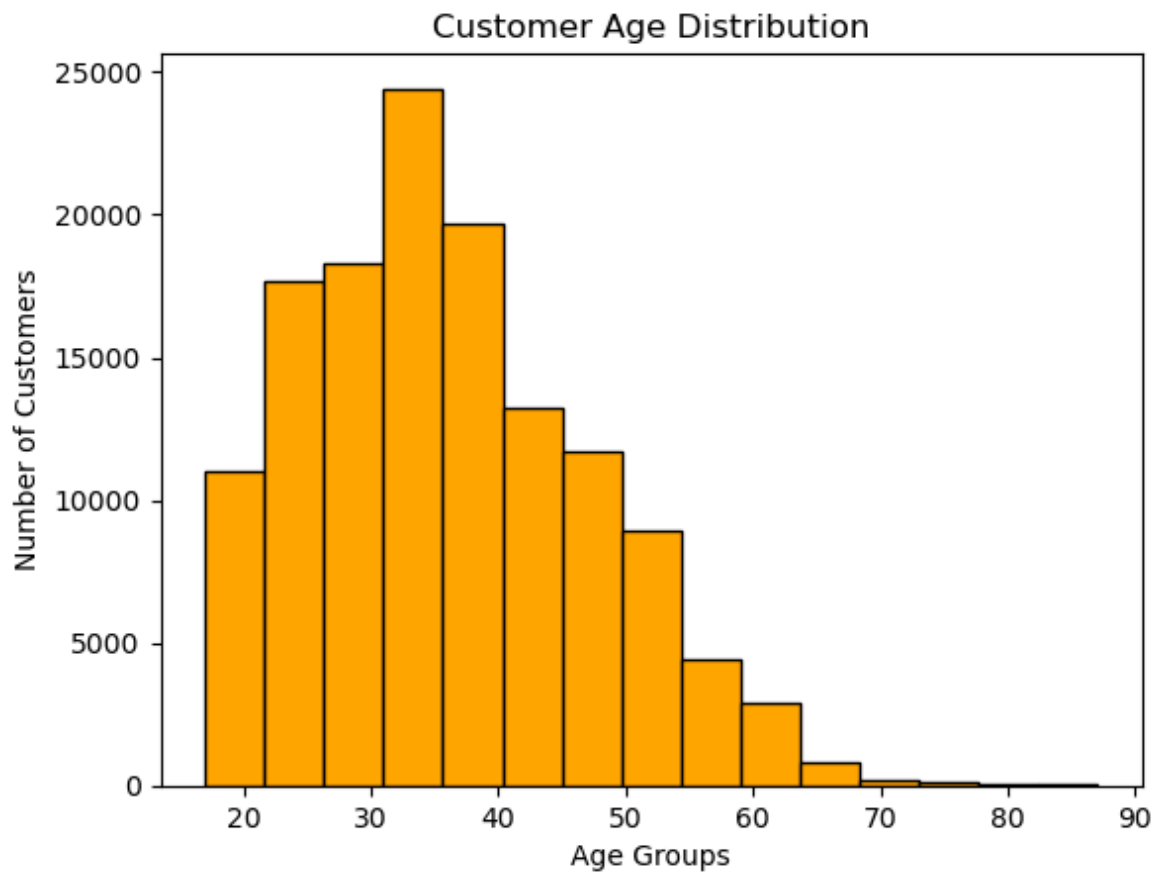


```
In [12]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
sub, cts = np.unique(pd.read_csv(r'C:\Users\khara\Downloads\Fliptkart Sales Datas
plt.pie(cts, labels=sub)
plt.show()
```



```
In [17]: import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv(r'C:\Users\khara\Downloads\Fliptkart Sales Dataset.csv')
plt.hist(df['Customer Age'], bins=15, color='orange', edgecolor='black')
plt.xlabel('Age Groups')
plt.ylabel('Number of Customers')
```

```
plt.title('Customer Age Distribution')  
plt.show()
```



In []: